Survival Analysis Project 1

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I. Introduction

HIV remains a major global public health challenge, with prevention continuing to play a critical role in reducing transmission. Recent advances have focused on antibody-based interventions, including VRC01, a broadly neutralizing antibody designed to target multiple HIV strains. While early studies suggest potential benefits, its effectiveness in preventing HIV infection must be evaluated through large-scale clinical trials. The primary objective of this analysis is to determine whether VRC01 reduces the risk of HIV infection, quantified through hazard ratios and examines if age and geographic region influence infection risk. Using Kaplan-Meier estimation and Cox proportional hazards models, this analysis aims to provide a clear assessment of treatment effectiveness and key determinants of HIV infection risk.

II. Methods:

The *amp.csv* dataset consists of several variables of interest, including *tx* (or *tx_pool*) as treatment variable; while *age*, *region1, region2*, *region3* and *region4* are confounders. The primary response variable is time to HIV-1 infection, defined by the indicator *hiv1event* and corresponding follow-up time *hiv1survday.*

Variable Name	Description
hiv1event	Response variable: Whether patient get HIV infected
tx	Whether patient receives high-dose, low-dose VRC01 treatment or placebo
tx_pool	Whether patient receives VRC01 treatment or placebo
hiv1survday	Follow-up time in days since enrollment
age	Patient's age
region_	4 dummy variables representing geographic region

Table 1: Variable name and description of each variable in the dataset

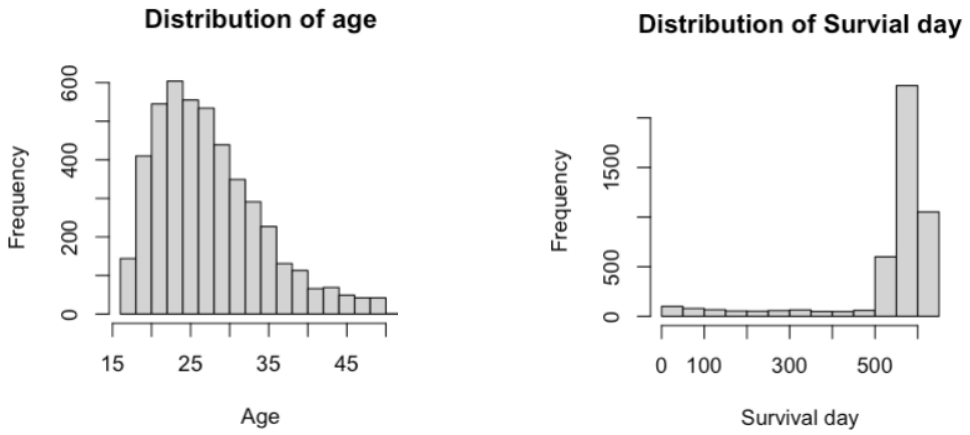
We analyzed data from two randomized controlled trials, HVTN 703 and HVTN 704, to evaluate the efficacy of VRC01 in preventing HIV infection. Variable *country* was omitted in the study because of the imbalance within each category data size. Then, the *tx_pool* variable was re-coded such that 0 indicated participants receiving treatment and 1 indicated those receiving placebo. Kaplan-Meier survival curves were first constructed by the treatment and placebo group across the entire follow-up time, followed by a log-rank test to assess their differences in the probability of HIV infection. In addition, we fitted a series of Cox proportional hazards models to determine whether these covariates influence HIV infection, including: (1) a model with only pooled treatment (VRC01 vs. placebo), (2) a model distinguishing low- and high-dose groups vs. placebo, (3) a multivariable model adjusting for age and geographic region, and (4) a model involving interaction terms between treatment and age to evaluate effect modification. Lastly, to quantify the treatment effectiveness, hazard ratios and 95% confidence intervals were estimated for all variables. All tests and analysis were performed using R-studio software.

III. Results

The descriptive statistics and distribution for the *hiv1survday* and *age* are outlined in Table 2 and Graph 1 & 2, while distribution of 4 geographic regions are shown in Table 3 and histogram figures of primary variable of interest is in Graph 3 & 4.

Variables	Mean	Median	Min	Max	Std.Dev.
hiv1survday	525.8	568	0	601	135.48
age	27.87	27	17	52	6.835

Table 2: Descriptive Statistics of numerical variables

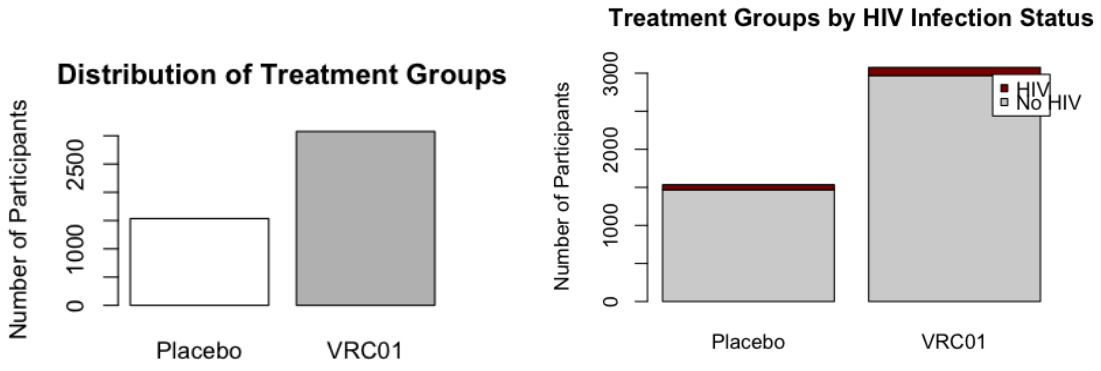


Graph 1&2: Distribution of age and Survival Day

	USA/Switzerland	Brazil/Peru	South Africa	Other Sub-Saharan Africa
Count	1413	1274	1019	905
Percentage	30.64%	27.63%	22.10%	19.63%

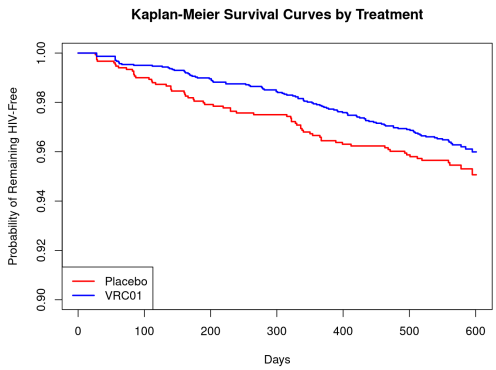
Table 3: Count and Percentage of 4 geographic regions recorded in study

Age distribution is right-skewed, with a mean of 27.87 years, while survival time distribution is left-skewed, with a mean of 525.8 days. Participants from the USA/Switzerland were the most common, accounting for 30.64% of the sample. Additionally, the number of participants receiving treatment (now tx_pool=1) was more than double that of those receiving placebo (tx_pool=0).



Graph 3&4: Number of participants and HIV infected people in among groups

Kaplan-Meier survival curves were conducted among treatment groups (VRCO1 vs. placebo) and we observed that two curves remain close together during the follow-up period.



Graph 5: Kaplan-Meler Survival curves by treatment groups

The log-rank test on *tx_pool* yielded $\chi^2 = 2.2$ with 1 degree of freedom, returning 0.10 p-value. After that a series of Cox proportional hazards models were fitted. The first model included treatment *tx_pool* only. Coefficient of *tx_pool* is -0.2304 then hazard ratio for pooled VRC01 versus placebo was 0.7942 with the p-value equivalent to 0.139 (95% CI: 0.5852-1.078).

In the second model we used *tx* to separate treatment into low-dose and high-dose groups. The low-dose group had a hazard ratio of 0.8909 with 0.516 p-value, while the high-dose group had a hazard ratio of 0.6976 with 0.058 p-value. P-values of both groups are greater than 0.05 significant level. The third model included again *tx* treatment (low- and high-dose), *age*, and geographic region, using *region4* as the reference group. The low-dose hazard ratio was 0.8867 (95% CI: 0.6259 - 1.2563, p = 0.499), and the high-dose hazard ratio was 0.6847 (95% CI: 0.4715-0.9943, p = 0.047). *Age* was associated with a hazard ratio of 0.9302 per year (95% CI: 0.9045-0.9566, p < 0.001). Geographic regions showed variation: *region1* had a hazard ratio of 0.5263 (95% CI: 0.3084-0.8981, p = 0.019), *region2* had a hazard ratio of 1.7036 (95% CI: 1.1306-2.5669, p = 0.011), and *region3* had a hazard ratio of 1.0246 (95% CI: 0.6496-1.6160, p

= 0.917). The fourth model meanwhile included interaction terms between treatment *tx_f* and *age*. However, neither predictor has a p-value lower than 0.05 significance level.

IV. Discussion

We found that that treatment with VRC01, whether pooled or separated into low-dose and high-dose groups, did not significantly reduce the risk of HIV infection. Visual check through Kaplan-Meier curves showed there is no significant difference between survival curves among treatment groups, while both the log-rank test and Cox proportional hazards models showed p-values above 0.05 for all treatment comparisons, suggesting that the treatment effect is not statistically distinguishable from placebo. Hazard ratios for treatment have confidence intervals including 1, indicating no robust evidence of treatment effectiveness. In contrast, age and geographic region were significantly associated with HIV infection. Older participants consistently showed a lower hazard of infection as hazard ratio per year is less than 1, while participants from different regions experienced varying risks. Specifically, individuals from the USA/Switzerland had a lower risk at HR=0.5263 relative to other sub-Saharan African countries, whereas those from Brazil/Peru had a higher risk at HR=1.7036. No significant difference was observed for participants from South Africa. Additionally, none of the interaction effects between treatment and *age* reached statistical significance, suggesting that *age* does not modify the effect of VRC01 on HIV infection.

Several limitations should be considered. First, the number of HIV infection events was relatively small, potentially limiting statistical power to detect modest treatment effects. Second, adherence to the intervention and other behavioral risk factors were not incorporated in these models, which could influence observed associations. Future research could explore larger or pooled datasets and investigate additional demographic or behavioral factors that may affect HIV

susceptibility. In short, VRC01 does not significantly reduce the overall risk of HIV infection in this study

V. Appendix

1. Data Exploration

```
library(survival)

library(survminer)

amp <- read.csv("amp.csv")



# check structure

summary(amp)

table(amp$hiv1event)

table(amp$tx)

table(amp$tx_pool)

table(amp$protocol)

table(amp$country)

summary(amp$hiv1survday)
```

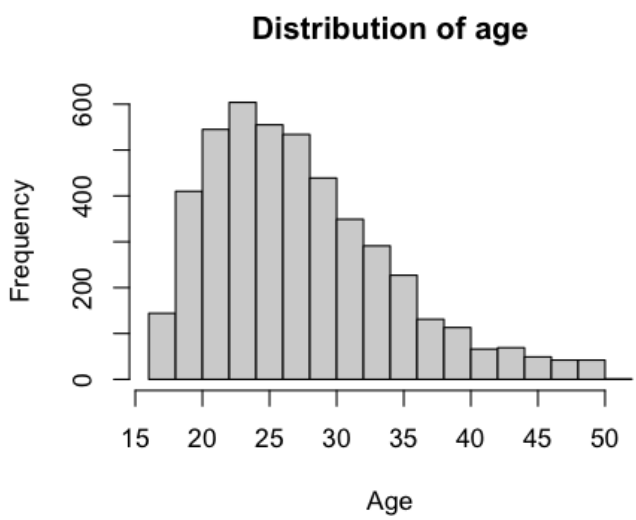
```
summary(amp$hiv1survday)
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  0.0   546.5   568.0   525.8   596.0   601.0 
```

```
summary(amp$age)
```

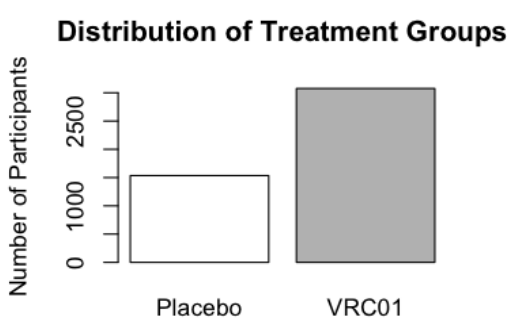
```
summary(amp$age)
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
17.00   23.00   27.00   27.87   32.00   52.00 
```

check distribution

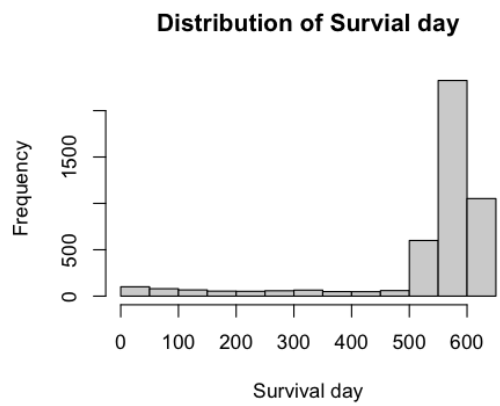
hist(amp$age, main="Distribution of age", xlab="Age")



hist(amp$tx, main="Distribution of Treatment Groups")



hist(amp$hiv1survday, main="Distribution of Survial day", xlab="Survival day")




```
region_pct <- region_count / nrow(amp) * 100

region_summary <- data.frame(

  Count = region_count,

  Percent = round(region_pct, 2)

)

region_summary
```

```
        Count Percent
Region1  1413   30.64
Region2  1274   27.63
Region3  1019   22.10
Region4   905   19.63
```

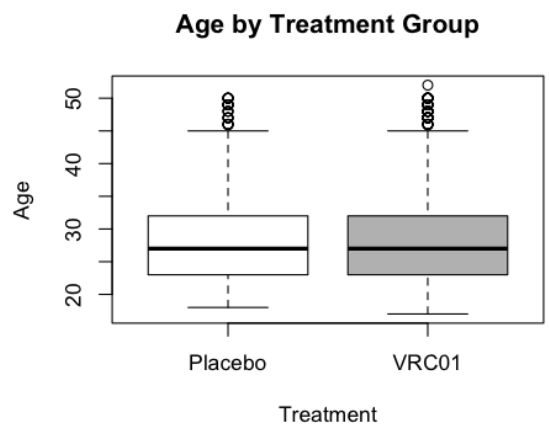
2. **Data Transformation**

```
#converts tx_pool, tx, protocol into categorical variables

amp$tx_pool_f <- factor(amp$tx_pool, levels = c(1, 0),labels = c("Placebo", "VRC01"))
```

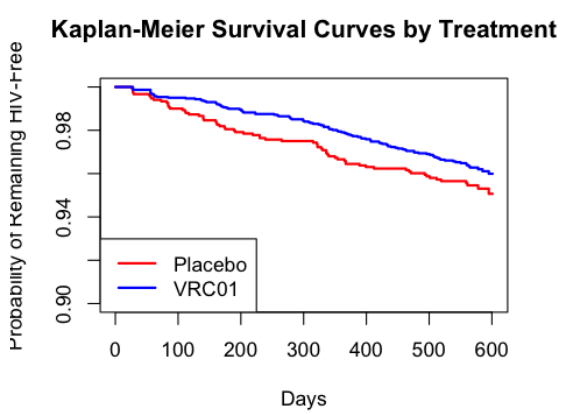
```
amp$tx_f <- factor(amp$tx,levels = c("C3", "T1", "T2"),labels = c("Placebo", "Low dose", "High dose"))

amp$protocol_f <- factor(amp$protocol)
```

3. Visual check

```
# box plot check between 2 treatment groups
boxplot(age ~ tx_pool_f,data = amp,main = "Age by Treatment Group",xlab = "Treatment",ylab = "Age",col = c("white", "grey"))
```




```
#kaplan meier curve for each treatment
km_fit <- survfit(Surv(hiv1survday, hiv1event) ~ tx_pool_f, data = amp)
plot(km_fit,col = c("red", "blue"),lwd = 2,xlab = "Days",ylab = "Probability of Remaining HIV-Free",main = "Kaplan-Meier Survival Curves by Treatment", ylim = c(0.9, 1))
```



4. Log-rank test

#log-rank test

survdiff(Surv(hiv1survday, hiv1event) ~ tx_pool_f, data = amp)

```
> survdiff(Surv(hiv1survday, hiv1event) ~ tx_pool_f, data = amp)
Call:
survdiff(formula = Surv(hiv1survday, hiv1event) ~ tx_pool_f,
    data = amp)

                     N Observed Expected (O-E)^2/E (O-E)^2/V
tx_pool_f=Placebo 1535       67     57.8     1.467       2.2
tx_pool_f=VRC01   3076      107    116.2     0.729       2.2

 Chisq= 2.2  on 1 degrees of freedom, p= 0.1
```

5. Cox proportional model

cox1 <- coxph(Surv(hiv1survday, hiv1event) ~ tx_pool_f, data = amp)

summary(cox1)

```
Call:
coxph(formula = Surv(hiv1survday, hiv1event) ~ tx_pool_f, data = amp)

  n= 4611, number of events= 174

                  coef exp(coef) se(coef)      z Pr(>|z|)
tx_pool_fVRC01 -0.2304    0.7942   0.1558 -1.479    0.139

               exp(coef) exp(-coef) lower .95 upper .95
tx_pool_fVRC01    0.7942      1.259    0.5852     1.078

Concordance= 0.529  (se = 0.019 )
Likelihood ratio test= 2.14  on 1 df,   p=0.1
Wald test            = 2.19  on 1 df,   p=0.1
Score (logrank) test = 2.2  on 1 df,   p=0.1
```

#fit the second Cox model (split treatment into high and low dose)

cox2 <- coxph(Surv(hiv1survday, hiv1event) ~ tx_f, data = amp)

summary(cox2)

```
Call:
coxph(formula = Surv(hiv1survday, hiv1event) ~ tx_f, data = amp)

  n= 4611, number of events= 174

                 coef exp(coef) se(coef)      z Pr(>|z|)
tx_fLow dose  -0.1155    0.8909   0.1777 -0.650   0.5158
tx_fHigh dose -0.3601    0.6976   0.1903 -1.893   0.0584 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

              exp(coef) exp(-coef) lower .95 upper .95
tx_fLow dose     0.8909      1.122    0.6288     1.262
tx_fHigh dose    0.6976      1.434    0.4804     1.013

Concordance= 0.542  (se = 0.02 )
Likelihood ratio test= 3.73  on 2 df,   p=0.2
Wald test            = 3.62  on 2 df,   p=0.2
Score (logrank) test = 3.66  on 2 df,   p=0.2
```

#fit the third Cox model ()

cox3 <- coxph(Surv(hiv1survday, hiv1event) ~ tx_f + age + region1 + region2 + region3,data = amp)

summary(cox3)

```
Call:
coxph(formula = Surv(hiv1survday, hiv1event) ~ tx_f + age + region1 +
    region2 + region3, data = amp)

  n= 4611, number of events= 174

                  coef exp(coef) se(coef)      z Pr(>|z|)
tx_fLow dose  -0.12023   0.88672  0.17776 -0.676   0.4988
tx_fHigh dose -0.37873   0.68473  0.19034 -1.990   0.0466 *
age           -0.07236   0.93020  0.01429 -5.063 4.12e-07 ***
region1       -0.64189   0.52630  0.27265 -2.354   0.0186 *
region2        0.53274   1.70359  0.20917  2.547   0.0109 *
region3        0.02428   1.02458  0.23248  0.104   0.9168
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

              exp(coef) exp(-coef) lower .95 upper .95
tx_fLow dose     0.8867      1.128    0.6259    1.2563
tx_fHigh dose    0.6847      1.460    0.4715    0.9943
age              0.9302      1.075    0.9045    0.9566
region1          0.5263      1.900    0.3084    0.8981
region2          1.7036      0.587    1.1306    2.5669
region3          1.0246      0.976    0.6496    1.6160

Concordance= 0.677  (se = 0.02 )
Likelihood ratio test= 72.24  on 6 df,   p=1e-13
Wald test            = 62.59  on 6 df,   p=1e-11
Score (logrank) test = 65.83  on 6 df,   p=3e-12
```

#interaction effect

cox_int <- coxph(Surv(hiv1survday, hiv1event) ~ tx_f+ tx_f * age,data = amp)

summary(cox_int)

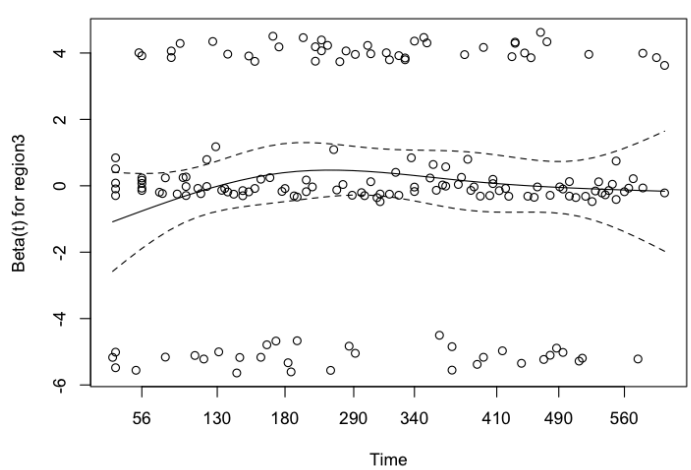
```
Call:
coxph(formula = Surv(hiv1survday, hiv1event) ~ tx_f + tx_f *
    age, data = amp)

  n= 4611, number of events= 174

                       coef exp(coef)  se(coef)      z Pr(>|z|)
tx_fLow dose      -0.205010  0.814639  0.851941 -0.241 0.809835
tx_fHigh dose      0.370605  1.448611  0.937785  0.395 0.692701
age               -0.076562  0.926295  0.022874 -3.347 0.000817 ***
tx_fLow dose:age   0.003229  1.003235  0.032956  0.098 0.921938
tx_fHigh dose:age -0.030460  0.970000  0.037329 -0.816 0.414509
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                  exp(coef) exp(-coef) lower .95 upper .95
tx_fLow dose         0.8146     1.2275    0.1534    4.3265
tx_fHigh dose        1.4486     0.6903    0.2305    9.1032
age                  0.9263     1.0796    0.8857    0.9688
tx_fLow dose:age     1.0032     0.9968    0.9405    1.0702
tx_fHigh dose:age    0.9700     1.0309    0.9016    1.0436

Concordance= 0.644  (se = 0.021 )
Likelihood ratio test= 44.7  on 5 df,   p=2e-08
Wald test            = 36.07  on 5 df,   p=9e-07
Score (logrank) test = 37.89  on 5 df,   p=4e-07
```

#ph assumptions of model 3

cox.zph(cox3)

```
> #ph assumptions of model 3
> cox.zph(cox3)
          chisq df     p
tx_f    5.40909  2 0.067
age     0.90870  1 0.340
region1 0.00219  1 0.963
region2 0.29211  1 0.589
region3 0.01829  1 0.892
GLOBAL  6.89271  6 0.331
```

plot(cox.zph(cox3))



The proportional hazards assumption for Model 3 was assessed using the Schoenfeld residuals test (cox.zph). No significant violations were detected for any covariate: treatment ($\chi^2 = 5.41$, df = 2, p = 0.067), age ($\chi^2 = 0.91$, df = 1, p = 0.340), region1 ($\chi^2 = 0.002$, df = 1, p = 0.963), region2 ($\chi^2 = 0.29$, df = 1, p = 0.589), and region3 ($\chi^2 = 0.018$, df = 1, p = 0.892), and the global test confirmed no overall violation ($\chi^2 = 6.89$, df = 6, p = 0.331). Visual inspection of the scaled Schoenfeld residual plots further supported this, showing no systematic trends over time. Therefore, the proportional hazards assumption was considered satisfied for all predictors in the model.